

Agenda

1. Friday Feedback
2. Section 9.3



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Mixing Problems

Mixing Problems refer to a collection of problems

where two or more solutions are mixed at various rates

Example: Suppose that a vat contains 100L of water initially. ^{$S(0) = 0$}
^{Fancy word for salt water} Brine enters the vat at a concentration of ^{the amount in vat is constant}
5g of salt per liter with a rate of 2L per minute $5 \frac{g}{L} \cdot 2 \frac{L}{min} = 10 \frac{g}{min}$

through pipe A. Brine also enters the vat at a
Concentration of 10g of salt per liter with a rate of $10 \frac{g}{L} \cdot 1 \frac{L}{min} = 10 \frac{g}{min}$
1L per minute through pipe B. The vat solution is ^{Concentration} of salt ^{is uniform}

Well-mixed. Brine leaves the vat at 3L per minute.

Find an equation $S(t)$ for the amount of salt in the
Vat at time t . ^{grams} _{minutes} $\frac{S(t)}{100} \frac{g}{L} \cdot 3 \frac{L}{min} = \frac{3S(t)}{100} \frac{g}{min}$

Equation in words:

Rate of change of salt = Salt Pipe A + Salt Pipe B - Salt leaving

$$\frac{ds}{dt} = 10g/min + 10g/min - \frac{3s}{100g/min}$$

dc

$$\frac{ds}{dt} = 20 \text{ g/m} - \frac{3s}{100} = \frac{2000}{100} - \frac{3s}{100}$$

Separate $\left(\frac{ds}{dt} = \frac{2000 - 3s}{100} \right)$

$$\int \frac{ds}{2000 - 3s} = \int \frac{dt}{100}$$

$u = 2000 - 3s$
 $du = -3ds$

$$= \frac{t}{100} + C$$

$$-\frac{1}{3} \int \frac{1}{u} du = \frac{t}{100} + C$$

$$-\frac{1}{3} \ln|u| = \frac{1}{100}t + C$$

$$-\frac{1}{3} \ln|2000 - 3s| = \frac{1}{100}t + C$$

$$e \cdot \ln|2000 - 3s| = \int \frac{-3}{100}t + D$$

$$|2000 - 3s| = e^{\left(\frac{-3}{100}t + D\right)} = e^{\frac{-3}{100}t} e^D$$

$$|2000 - 3s| = E e^{\frac{-3}{100}t} \quad E = e^D \text{ (note } E > 0)$$

$$F = \pm E$$

$$-3s = F e^{\frac{-3}{100}t} - 2000$$

$$s = \frac{F}{-3} e^{\frac{-3}{100}t} + 2000/3$$

$$G = \underline{F}$$

$$s(t) = G e^{\frac{-3}{100}t} + \frac{2000}{3}$$

$$s(0) = G e^{\frac{-3}{100}(0)} + \frac{2000}{3}$$

$$s(0) = 2000$$

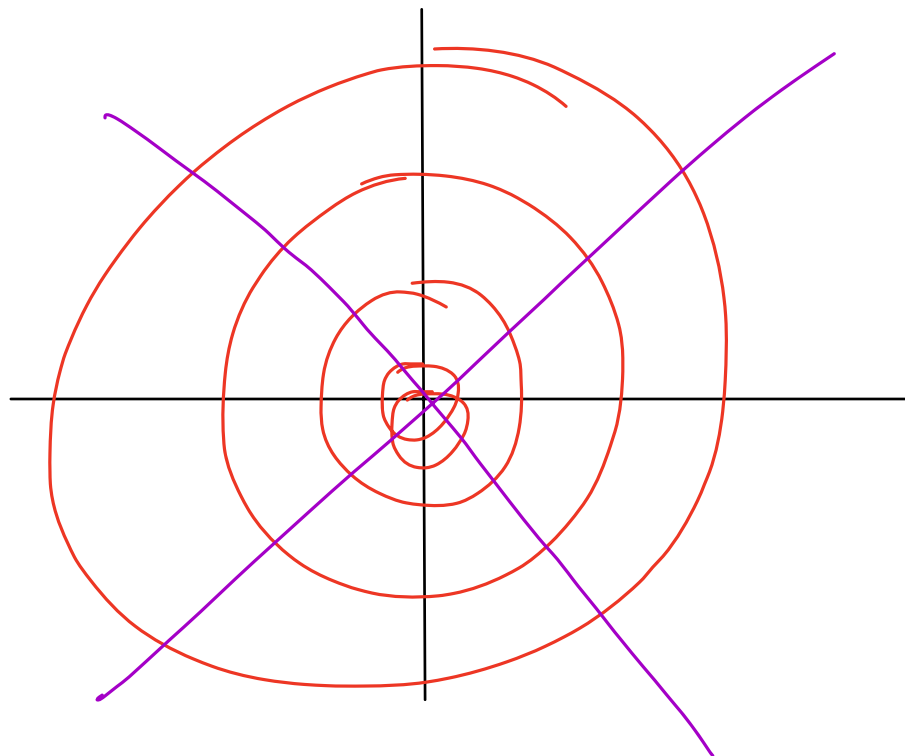
$$s(t) = \frac{-2000 \cdot 3}{3} e^{\frac{-3}{100}t} + \frac{2000}{3}$$

$$G = -\frac{2000}{3}$$

Orthogonal Trajectories

An orthogonal trajectory is a curve that intersects all curves in a family of curves orthogonally

Example:



Example: Find the orthogonal trajectories of the curves $xy = k \leftarrow y = \frac{k}{x}$ where k varies.

① Find y' : $-kx^{-2} = -\frac{k}{x^2}$

② ^{remove} k = $-\frac{xy}{x^2} = \left(\frac{-y}{x}\right)$

notice $k = xy$

Slopes of original family of curves

③ set up differ. representing

$$\frac{dy}{dx} = \frac{-1}{\frac{-y}{x}} = \left(\frac{x}{y} \right)$$

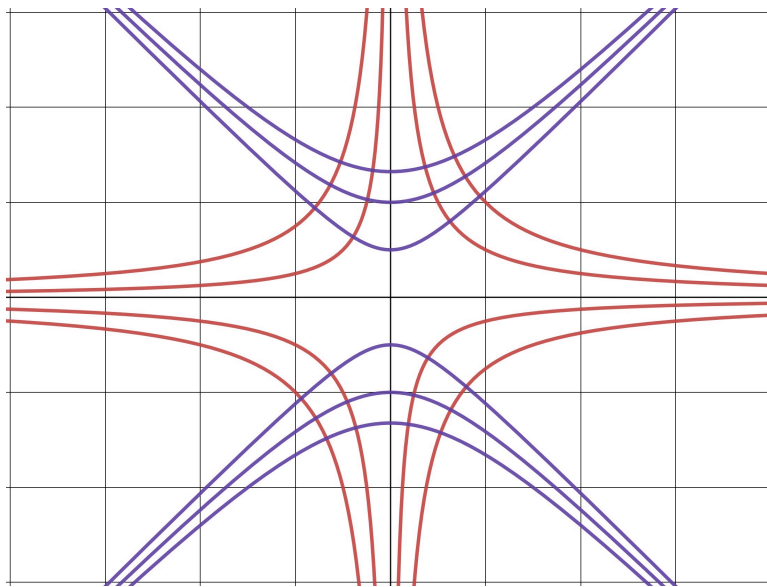
Slopes of orthogonal trajectories.

④ solve differ. in ③

$$\int y dy = \int x dx$$

$$\frac{1}{2} y^2 = \frac{1}{2} x^2 + C$$

$$y^2 = x^2 + C \Rightarrow y^2 + x^2 = C$$



original fam. of curves $\left(y = \frac{k}{x} \right)$

$$y^2 + x^2 = C$$

$$\begin{array}{r} 0111.10 \\ + 0000.11 \\ \hline \end{array}$$

0000,01

two $(+) \neq (-) \rightarrow$ overflow

carry
 11
 1010.10
 + 1110.00

1000.10

Still $\hookrightarrow$ so no overton